

Group-Balanced Counterfactual Logit Pairing for Robust Toxicity Detection under Demographic Subpopulation Shift

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Abstract

Toxicity detection models are widely used in online moderation, but strong average performance can hide uneven behaviour across demographic subgroups and language varieties. This paper studies whether identity-term debiasing methods remain reliable under demographic subpopulation shift and whether their fairness gains transfer to a dialect-oriented evaluation setting. Building on prior work on unintended bias and counterfactual fairness, I evaluate four methods on CivilComments-WILDS: empirical risk minimisation (ERM), counterfactual data augmentation (CDA), counterfactual logit pairing (CLP), and a proposed group-balanced counterfactual logit pairing method (GB-CLP). The proposed method combines identity-term counterfactuals with inverse-frequency weighting over label-identity groups. Experiments use a DistilBERT toxicity classifier, official CivilComments-WILDS splits, validation-based checkpoint selection, and validation-only threshold tuning. Results show that the proposed method does not dominate all metrics, but it reduces the maximum absolute false-positive-rate gap on CivilComments from 0.1151 for ERM to 0.0807, while keeping macro-F1 close to the strongest baseline. However, CLP achieves the best worst-group accuracy, ERM achieves the highest ROC-AUC and worst-group AUC, and CDA performs best on the exploratory dialect-transfer false-positive-rate gap. These findings suggest that counterfactual debiasing can change fairness behaviour under identity-based evaluation, but the benefits are metric-dependent and should not be assumed to transfer automatically to dialectal variation.

Keywords

toxicity detection, unintended bias, counterfactual logit pairing, subpopulation shift, fairness, dialectal variation

1 Introduction

Automatic toxicity detection is often used to support large-scale content moderation, where platforms need to process more comments than human moderators can review manually. Prior work has shown that these systems can perform well on average while still behaving unfairly for comments that mention particular demographic identities [2, 5]. This creates a practical problem because false positives can suppress non-toxic discussion, while false negatives can leave harmful comments visible.

The limitation studied in this paper is not simply that toxicity classifiers can be biased. That issue is already well established. The more specific question is whether debiasing methods remain reliable when the evaluation setting becomes harder. In particular, this work focuses on demographic subpopulation shift and dialectal variation. CivilComments-WILDS is suitable for this setting because it treats identity-defined groups as subpopulations and evaluates

robustness under distribution shift [7]. Dialectal variation is also important because prior work has shown that African American English (AAE)-aligned text can be treated differently by hate-speech and toxicity systems [1, 9].

This paper proposes and evaluates *Group-Balanced Counterfactual Logit Pairing* (GB-CLP). The method extends counterfactual logit pairing [6] by adding inverse-frequency weights over label-identity groups. The aim is to reduce sensitivity to identity-term substitutions while also increasing the influence of underrepresented label-identity combinations during training.

The contribution of this paper is empirical and diagnostic. I compare GB-CLP against ERM, CDA, and CLP using the same model, tokenizer, dataset splits, training budget, validation protocol, and thresholding procedure. The evaluation reports both standard predictive metrics and fairness-oriented metrics, including worst-group accuracy, worst-group AUC, Subgroup AUC, BPSN AUC, BNSP AUC, and false-positive-rate gaps. The results show a clear fairness-performance trade-off rather than a simple win for one method.

2 Related Work

Toxicity detection has been studied as a large-scale text classification problem, with early work showing how crowdsourcing and machine learning can be used to identify personal attacks at scale [10]. Later studies showed that toxicity classifiers can learn unintended associations between demographic identity terms and toxicity labels. Dixon et al. [5] examined unintended bias in text classification, while Borkan et al. [2] introduced Subgroup AUC, BPSN AUC, and BNSP AUC to measure different forms of identity-related bias.

Counterfactual debiasing directly targets prediction changes caused by identity-term substitutions. Garg et al. [6] proposed counterfactual data augmentation and counterfactual logit pairing, which encourage models to behave more consistently when demographic terms are swapped. These methods are useful because they provide an explicit training signal for identity-term invariance, but they may not improve every group-fairness metric or transfer to dialectal variation.

Subpopulation shift provides a more demanding evaluation setting. WILDS was introduced as a benchmark for distribution shifts that occur in real-world applications, and CivilComments-WILDS is relevant here because demographic identities define subpopulations and evaluation emphasises robustness across those groups [7]. Dialectal bias is related but distinct. Blodgett et al. [1] developed a demographic dialect framework for studying AAE-aligned language, and Sap et al. [9] showed that hate-speech detection systems can reflect racialised dialect bias. Davidson et al. [3] provide

the hate-speech and offensive-language dataset used in this paper for external transfer evaluation, although the present study uses only an exploratory proxy grouping rather than a validated dialect classifier.

3 Preliminaries and Problem Definition

Let x_i be a comment, $y_i \in \{0, 1\}$ be its toxicity label, and $f_\theta(x_i)$ be the model logit. The predicted toxicity probability is

$$p_\theta(y_i = 1 | x_i) = \sigma(f_\theta(x_i)), \quad (1)$$

where $\sigma(\cdot)$ is the sigmoid function. A standard toxicity classifier can be trained using binary cross-entropy:

$$\mathcal{L}_{\text{BCE}}(x_i, y_i) = -y_i \log p_i - (1 - y_i) \log(1 - p_i). \quad (2)$$

The main problem is to reduce unfair behaviour under demographic subpopulation shift without losing too much overall predictive quality. In CivilComments-WILDS, identity metadata define demographic subgroups. In this paper, the group used for training weights is

$$g_i = (y_i, a_i), \quad (3)$$

where a_i indicates whether the comment contains at least one tracked identity mention. This creates four label-identity groups: non-toxic without identity, non-toxic with identity, toxic without identity, and toxic with identity.

The fairness question is evaluated using both threshold-independent and threshold-dependent measures. Threshold-independent metrics include ROC-AUC, Subgroup AUC, BPSN AUC, BNPS AUC, and worst-group AUC. Threshold-dependent metrics include macro-F1, worst-group accuracy, and false-positive-rate gaps. The threshold is selected only on the validation set using macro-F1, then fixed for test evaluation.

4 Proposed Method

The proposed method, GB-CLP, combines two ideas. The first is counterfactual logit pairing. For a comment x_i , an identity-swapped counterfactual x'_i is generated by replacing explicit identity terms using transparent swap pairs, such as *muslim/christian*, *black/white*, and *woman/man*. The CLP loss encourages the original and counterfactual logits to remain close:

$$\mathcal{L}_{\text{pair}}(x_i, x'_i) = (f_\theta(x_i) - f_\theta(x'_i))^2. \quad (4)$$

The second idea is group-balanced weighting. Let n_g be the number of training examples in group g . Each training example receives a normalised inverse-frequency weight w_{g_i} , so that smaller label-identity groups have greater influence during optimisation. This is intended to reduce the chance that the model is dominated by the largest group.

For examples where the counterfactual text was actually changed, the proposed loss is

$$\begin{aligned} \mathcal{L}_{\text{GB-CLP}} = & \frac{1}{N} \sum_{i=1}^N w_{g_i} [\mathcal{L}_{\text{BCE}}(x_i, y_i) + \mathcal{L}_{\text{BCE}}(x'_i, y_i)] \\ & + \lambda \frac{\sum_{i=1}^N m_i \mathcal{L}_{\text{pair}}(x_i, x'_i)}{\sum_{i=1}^N m_i}. \end{aligned} \quad (5)$$

where $m_i = 1$ if the identity swap changed the text and $m_i = 0$ otherwise. In the experiments, $\lambda = 1.0$. This design addresses the

research problem by testing whether identity-invariant training, when combined with group balancing, improves robustness under subpopulation shift.

The method is intentionally simple. This is useful for controlled comparison, but it also creates limitations. Identity swapping can produce unnatural or socially simplified text, and explicit identity substitutions do not capture all forms of demographic or dialectal variation.

5 Experimental Setup

5.1 Datasets

The main benchmark is CivilComments-WILDS [7]. The experiment uses the official train, validation, and test splits rather than a random split because the goal is to preserve the benchmark’s subpopulation-shift structure. Controlled subsets were sampled from the official splits: 50,000 training examples, 10,000 validation examples, and 20,000 test examples. Sampling was stratified by label and whether any tracked identity was present.

For dialect-transfer evaluation, the Davidson hate-speech and offensive-language dataset was used as an external evaluation dataset [3]. Hate-speech and offensive-language labels were mapped to the toxic class, while the neither class was mapped to the non-toxic class. The notebook used 10,000 examples. Because no validated dialect probability file was used in the current experiment, the dialect analysis is reported only as an exploratory proxy evaluation. The two groups are labelled *AAE_proxy* and *Other_proxy*; they should not be interpreted as confirmed dialect labels.

5.2 Baselines and Model

All methods use *distilbert-base-uncased*, a distilled version of BERT [4, 8]. The four compared methods are:

- **ERM**: standard empirical risk minimisation with binary cross-entropy.
- **CDA**: counterfactual data augmentation using identity-swapped examples.
- **CLP**: counterfactual logit pairing with original and identity-swapped examples.
- **GB-CLP**: CLP with inverse-frequency label-identity group weighting.

All methods used the same model backbone, tokenizer, maximum sequence length, optimiser, validation protocol, and thresholding procedure. Models were trained for up to 10 epochs with early stopping. The checkpoint with the highest validation ROC-AUC was selected for final test evaluation. The classification threshold was then chosen on the validation set by maximising macro-F1 and was not tuned on the test set.

5.3 Evaluation Metrics

Overall predictive performance is measured using ROC-AUC, macro-F1, accuracy, precision, and recall. Fairness and robustness are measured using worst-group accuracy, worst-group AUC, mean Subgroup AUC, mean BPSN AUC, mean BNPS AUC, and the maximum absolute false-positive-rate gap across identity groups. Subgroup AUC, BPSN AUC, and BNPS AUC follow the unintended-bias evaluation framework introduced by Borkan et al. [2].

6 Results

6.1 CivilComments-WILDS Results

Table 1 reports the main CivilComments test results. The proposed method does not achieve the best value on every metric. ERM has the highest ROC-AUC at 0.9382 and the highest worst-group AUC at 0.8342. CLP has the best worst-group accuracy at 0.6416. The proposed method has the highest accuracy, although the difference from ERM is small, and it achieves the lowest maximum absolute false-positive-rate gap at 0.0807. This suggests that GB-CLP mainly improves the FPR-disparity measure rather than producing a universal improvement across all robustness metrics.

Figure 1 shows that all methods remain far above the random baseline and have similar ROC curves. This supports the table result that the debiasing methods did not severely damage overall ranking performance, although ERM remains slightly ahead in ROC-AUC.

The FPR-gap result in Figure 2 is the clearest positive result for the proposed method. Compared with ERM, GB-CLP reduces the maximum absolute FPR gap from 0.1151 to 0.0807. Compared with CLP, the reduction is larger, from 0.1892 to 0.0807. The BPSN result in Figure 3 is also favourable for GB-CLP, with mean BPSN AUC increasing to 0.8916. However, the lower BNSP AUC for GB-CLP shows that this improvement is not uniform across all unintended-bias metrics.

Figures 4 and 5 show why the result should be interpreted carefully. CLP improves worst-group accuracy, but it also produces the largest FPR gap. ERM has the best worst-group AUC, while the proposed method gives the lowest FPR gap. Therefore, the experiment does not support a broad claim that one debiasing method is best in all senses. Instead, the evidence suggests that different fairness metrics capture different behaviours.

6.2 Training Behaviour

The selected validation checkpoints were early in training: epoch 2 for ERM, epoch 1 for CDA, epoch 2 for CLP, and epoch 2 for the proposed method. Training continued for later epochs until early stopping, but validation ROC-AUC did not consistently improve after these points. This supports the use of validation-based checkpoint selection rather than simply using the final epoch.

6.3 Dialect-Transfer Evaluation

Table 2 reports the external Davidson transfer results. The dialect grouping is an exploratory proxy, so the results should be interpreted as a stress test rather than a validated measurement of AAE or White-aligned language. In this setting, ERM has the highest ROC-AUC and macro-F1. CDA gives the lowest proxy dialect FPR gap at 0.0171. The proposed method has a proxy FPR gap of 0.1061, close to ERM and CLP. This indicates that the proposed CivilComments FPR-gap improvement does not clearly transfer to the dialect-proxy setting.

Figure 6 supports the main generalisation finding. The proposed method improves one CivilComments fairness measure, but it does not automatically improve the proxy dialect-transfer gap. This is consistent with the concern that identity-term debiasing and dialectal robustness are related but not identical problems.

6.4 Error Analysis

Table 3 summarises the proposed method’s errors by primary identity group. The highest false-positive rate occurs for the *white* group at 0.1037. The highest false-negative rate occurs for the *LGBTQ* group at 0.1502, followed by *black* at 0.1212 and *muslim* at 0.1124. The no-identity group has a comparatively low false-negative rate of 0.0191, although it still contains many total errors because it is the largest group. This shows that aggregate accuracy can hide different error patterns across subgroups.

Manual inspection of high-confidence false positives and false negatives also suggests that some errors are not caused only by explicit identity terms. Several high-confidence false positives contain insults or aggressive language without tracked identity mentions, while several false negatives contain sarcastic, indirect, or context-dependent toxicity. This matters because counterfactual identity swapping mainly targets explicit identity terms and is less able to address pragmatic or discourse-level toxicity.

7 Discussion

The main finding is that GB-CLP improves some fairness measures but not all of them. On CivilComments-WILDS, it produces the lowest maximum absolute FPR gap and the highest mean BPSN AUC. This provides evidence that group-balanced counterfactual training can reduce one form of identity-related false-positive disparity in this controlled setting. However, the method has lower worst-group accuracy than ERM and CLP, and lower worst-group AUC than ERM. Therefore, GB-CLP should be interpreted as improving a specific fairness trade-off rather than as a uniformly stronger model.

The results also show that counterfactual debiasing can behave differently depending on the metric. CLP improves worst-group accuracy but increases the maximum absolute FPR gap. CDA performs worse than ERM on the main CivilComments metrics but gives the lowest proxy dialect-transfer FPR gap. These differences suggest that fairness evaluation should not rely on a single number.

The dialect-transfer result is especially important for interpreting generalisation. GB-CLP was designed around explicit identity-term counterfactuals and label-identity group balancing, while the Davidson proxy evaluation reflects a different source of variation. Since GB-CLP does not clearly improve the proxy dialect FPR gap, the experiment supports a limited conclusion: identity-term debiasing may help with some identity-based fairness measures, but it should not be assumed to solve dialectal robustness without direct evidence from a validated dialect evaluation.

From a practical moderation perspective, these results suggest that model selection depends on the type of harm a system is trying to reduce. If the priority is reducing false-positive-rate disparity across CivilComments identity groups, GB-CLP is the strongest option in this experiment because it gives the lowest maximum absolute FPR gap while keeping macro-F1 close to ERM. However, if the priority is worst-group accuracy or overall ranking performance, CLP or ERM may be preferable. The Davidson transfer result also suggests that identity-term debiasing should not be treated as dialect-bias mitigation without a separate validated dialect evaluation.

Table 1: CivilComments-WILDS test results. Higher is better for ROC-AUC, macro-F1, accuracy, worst-group accuracy, worst-group AUC, Subgroup AUC, BPSN AUC, and BNSP AUC. Lower is better for maximum absolute FPR gap.

Method	ROC-AUC	Macro-F1	Acc.	Worst Acc.	Worst AUC	Subgrp AUC	BPSN AUC	BNSP AUC	Max $ \Delta\text{FPR} $
ERM	0.9382	0.8002	0.9187	0.6168	0.8342	0.8754	0.8783	0.9403	0.1151
CDA	0.9323	0.7912	0.9184	0.5701	0.8046	0.8663	0.8595	0.9387	0.1350
CLP	0.9362	0.8001	0.9166	0.6416	0.8209	0.8717	0.8632	0.9438	0.1892
GB-CLP	0.9365	0.7990	0.9190	0.5599	0.8224	0.8731	0.8916	0.9287	0.0807

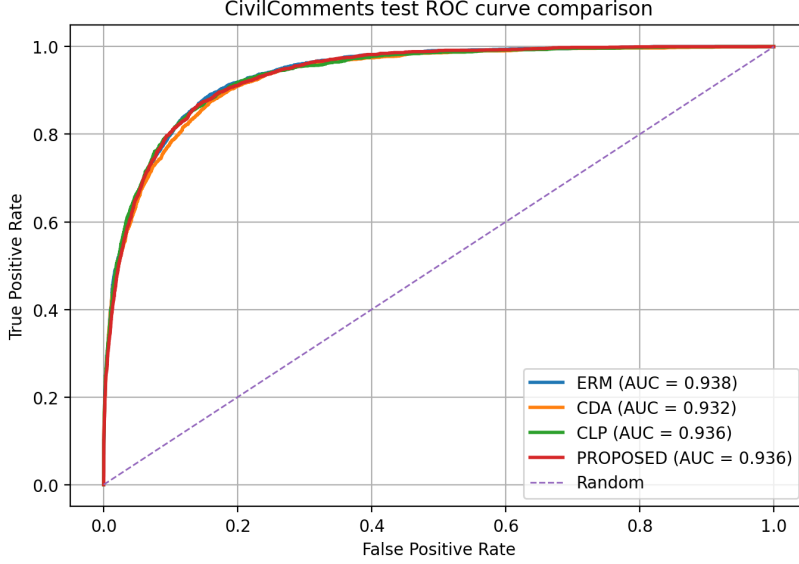


Figure 1: ROC curve comparison on the CivilComments-WILDS test subset. The curves are close, showing that the debiasing methods maintain broadly similar ranking performance to ERM.

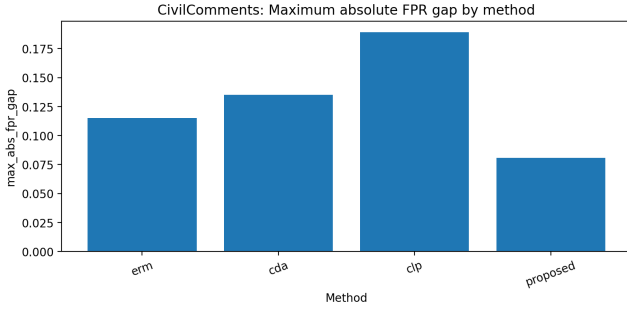


Figure 2: Maximum absolute false-positive-rate gap on CivilComments-WILDS. The proposed method gives the lowest FPR gap among the four methods.

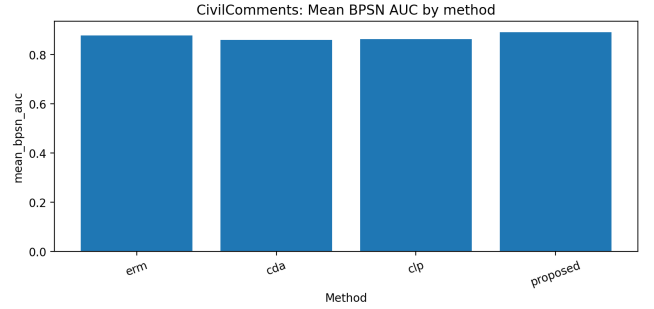


Figure 3: Mean BPSN AUC on CivilComments-WILDS. The proposed method has the highest BPSN AUC, suggesting better separation between non-toxic subgroup examples and toxic background examples.

8 Limitations

This study has several limitations. First, the experiment uses a controlled subset of CivilComments-WILDS rather than the full dataset, and the reported results come from one random seed. Therefore, the results should be interpreted as controlled empirical evidence rather

than a complete benchmark claim, and no statistical significance or confidence intervals are claimed. Second, the dialect-transfer analysis uses an exploratory lexical proxy rather than a validated dialect

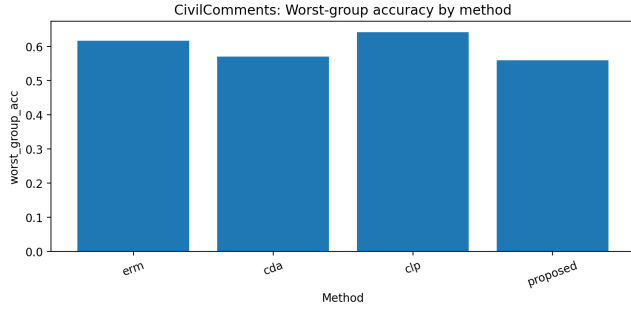


Figure 4: Worst-group accuracy on CivilComments-WILDS. CLP performs best on this metric, while the proposed method is lower despite its stronger FPR-gap result.

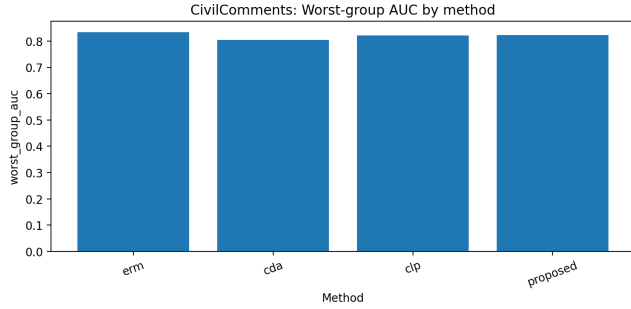


Figure 5: Worst-group AUC across identity subgroups. ERM achieves the highest value on this threshold-independent metric.

Table 2: Davidson transfer evaluation using exploratory proxy dialect groups. Higher is better for ROC-AUC and macro-F1. Lower is better for FPR gap.

Method	ROC-AUC	Macro-F1	Proxy FPR gap
ERM	0.8593	0.7031	0.1115
CDA	0.8456	0.6867	0.0171
CLP	0.8288	0.6861	0.1031
GB-CLP	0.8333	0.6825	0.1061

model, so the dialect results should not be treated as a definitive evaluation of AAE robustness.

Third, the identity-swapping procedure is transparent but simple. Some substitutions may create unnatural text or remove important social context, and explicit swaps do not capture all forms of demographic or dialectal variation. Fourth, all methods use the same DistilBERT backbone. This improves experimental control, but it does not show whether the same trends would hold for larger models or more recent language encoders. Finally, the proposed method balances only label-identity groups based on whether any tracked identity is present, rather than directly balancing every individual identity group during training.

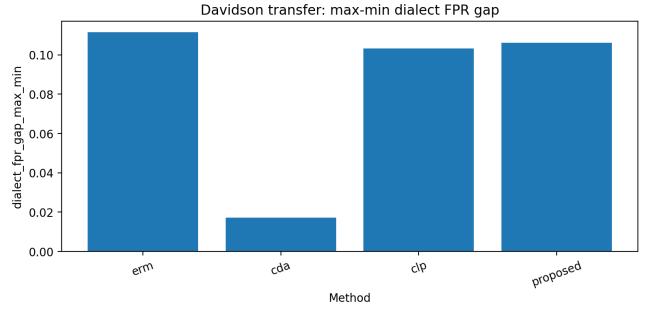


Figure 6: Exploratory Davidson transfer false-positive-rate gap. CDA gives the lowest proxy dialect gap, while the proposed method does not clearly transfer its CivilComments FPR-gap improvement.

Table 3: Error analysis for the proposed method by primary identity group.

Group	Total	FP rate	FN rate
LGBTQ	486	0.0432	0.1502
black	495	0.0566	0.1212
christian	1636	0.0263	0.0373
female	1642	0.0512	0.0591
male	2121	0.0434	0.0693
muslim	676	0.0444	0.1124
no identity	12123	0.0365	0.0191
other religions	252	0.0357	0.0635
white	569	0.1037	0.0896

9 Conclusion and Future Work

This paper evaluated whether debiasing methods for toxicity detection remain effective under demographic subpopulation shift and whether their gains transfer to a dialect-oriented evaluation setting. The proposed GB-CLP method combines counterfactual logit pairing with inverse-frequency label-identity group weighting. On CivilComments-WILDS, GB-CLP reduced the maximum absolute FPR gap and improved mean BPSN AUC while keeping macro-F1 close to the strongest baseline. However, ERM retained the best ROC-AUC and worst-group AUC, CLP achieved the best worst-group accuracy, and CDA performed best on the exploratory Davidson proxy FPR gap. These results indicate a fairness-performance trade-off rather than a single dominant method.

The main conclusion is therefore cautious. Group-balanced counterfactual training can improve selected fairness measures in an identity-based subpopulation-shift setting, but the improvement is metric-dependent and does not automatically generalise to the exploratory dialect-transfer setting. Future work should repeat the experiment with multiple random seeds, use the full CivilComments-WILDS dataset, include a validated dialect probability model, and test whether the same trade-offs hold for larger transformer backbones. It would also be useful to design counterfactuals that preserve more linguistic and social context rather than relying only on simple identity-term swaps.

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